

1. Find the z-transform and the ROC for the following sequences

- $x[n] = \left(\frac{1}{3}\right)^n u[-n]$
- $x[n] = \left(\frac{1}{3}\right)^n (u[n] - u[n-5])$
- $x[n] = n u[n] - (n-N)u[n-N]$
- $x[n] = a^{|n|} \quad 0 \leq |a| < 1$
- $x[n] = \begin{cases} 1 & 0 \leq n \leq N-1 \\ 0 & n \geq N \\ 0 & n < 0 \end{cases}$

1. a) $x[n] = \left(\frac{1}{3}\right)^n u[-n]$

$u[-n] = \begin{cases} 1 & -n \geq 0 \rightarrow n \leq 0 \\ 0 & -n < 0 \rightarrow n > 0 \end{cases}$

$$X(z) = \sum_{n=-\infty}^{+\infty} x[n] z^{-n} = \sum_{n=-\infty}^{+\infty} \left(\frac{1}{3}\right)^n u[-n] z^{-n} = \sum_{n=-\infty}^0 \left(\frac{1}{3}\right)^n z^{-n}$$

$$= \sum_{n=-\infty}^0 \left(\frac{1}{3} z^{-1}\right)^n = \sum_{q=0}^{\infty} \left(\frac{1}{3z}\right)^q = \sum_{q=0}^{\infty} \left(\frac{1}{3z}\right)^q$$

$$= \frac{1}{1 - \frac{1}{3z}} \quad \text{for } |3z| < 1 \rightarrow |z| < \frac{1}{3}$$

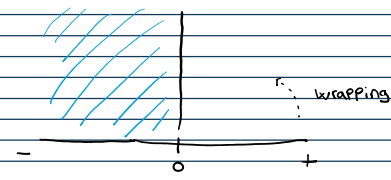
ROC:

Husk fra bogen.

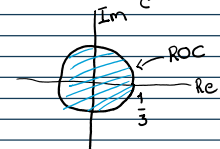
$$u[n] = \begin{cases} 1, & n \geq 0 \\ 0, & n < 0 \end{cases}$$

Flipper det bare
Så det står på formen
Ved derfor at det også er
på left-sided

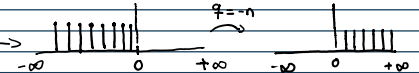
left-sided
ROC



wrapping
positive
rundt om negative



Flipper nu fra $-\infty \rightarrow 0$
til $0 \rightarrow +\infty$. Det kan ses lidt
som de negative frekvenser
der bliver spejlet over til
de positive.



$$0 \leq n \leq N-1 = 0 \leq n \leq 5-1$$

b) $x[n] = \left(\frac{1}{3}\right)^n (u[n] - u[n-5]) = \text{finite length sequence}$

$$u[n] = \begin{cases} 1 & n \geq 0 \\ 0 & n < 0 \end{cases} \quad u[n-5] = \begin{cases} 1 & n \geq 5 \\ 0 & n < 5 \end{cases}$$

$$\rightarrow u[n] - u[n-5] = \begin{cases} 1 & 0 \leq n \leq 4 \\ 0 & n < 0, n \geq 5 \end{cases}$$

$$X(z) = \sum_{n=-\infty}^{+\infty} \left(\frac{1}{3}\right)^n (u[n] - u[n-5]) z^{-n} = \sum_{n=0}^4 \left(\frac{1}{3}\right)^n z^{-n} = \sum_{n=0}^4 \left(\frac{1}{3z}\right)^n$$

$$\left| \sum_{n=0}^{N-1} a^n = \frac{1-a^N}{1-a} \quad |a| < 1 \right\} \text{Smart gili formel jeg ikke helt kan finde i bogen}$$

$$= \frac{1 - \left(\frac{1}{3z}\right)^5}{1 - \frac{1}{3z}} = \frac{1 - (3z)^{-5}}{1 - (3z)^{-1}} \quad |z| > 0$$

I finite length er vi lidt

"ligeglad" med unit step delen

Når vi laver z-transformen.

De fortæller bare vores boundary

conditions for vores z-transform

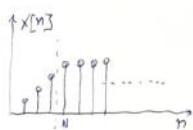
af a^n .

$$\sum_{n=-\infty}^{+\infty} x[n] \cdot z^{-n} \quad (\text{z-transform})$$

$$\sum_{n=0}^{N-1} a^n \cdot z^{-n} \quad (\text{z-transform af } a^n, \text{ aka finite length seq})$$

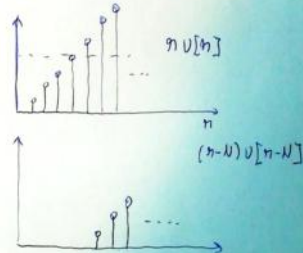
Man kunne også have anvendt
Sævnens (13) fra tabellen og få
det samme svar

c) $x[n] = \begin{cases} n & 0 \leq n \leq N-1 \\ N & n \geq N \end{cases}$



We observe that

$$x[n] = n u[n] - (n-N) u[n-N]$$



From the table of z-transform properties we know that

$$n x[n] \xrightarrow{z} -z \frac{d}{dz} X(z)$$

$$\text{Since } u[n] \xrightarrow{z} \frac{1}{1-z^{-1}}, \quad nu[n] \xrightarrow{z} -z \frac{d}{dz} \frac{1}{1-z^{-1}} = \frac{z^{-1}}{(1-z^{-1})^2} \quad |z| > 1$$

We also know that

$$x[n-n_0] \xrightarrow{z} X(z) z^{-n_0}$$

$$(n-N)u[n-N] \xrightarrow{z} \frac{z^{-1}}{(1-z^{-1})^2} \cdot z^{-N} = \frac{z^{-N-1}}{(1-z^{-1})^2} \quad |z| > 1$$

Therefore

$$x[n] = nu[n] - (n-N)u[n-N] \xrightarrow{z} X(z) = \frac{z^{-1} - z^{-N-1}}{(1-z^{-1})^2} = \frac{z^{-1}(1-z^{-N})}{(1-z^{-1})^2}$$

Hvis man ikke gider at gøre det på Gili's metode, og heller vil gøre det med tabellen, så gælder det om at indse nogle ting.

Sekvensen bliver delt op i 3 forskellige bidder.

1. $nu[n]$: ved at bruge sekvens (7)

$$(na^n u[n]) \xrightarrow{z} \frac{az^{-1}}{(1-az^{-1})^2} \text{ og så sætter } a=1$$

$$1 \text{ så får man: } n \cdot 1^n \cdot u[n] \xrightarrow{z} \frac{1 \cdot z^{-1}}{(1-z^{-1})^2} \rightarrow (1-z^{-1})^2 = 0 \Rightarrow z=1 \quad |z| > 1$$

2. $(n-N)$: det samme som før

$$(n-N) \cdot 1^n \cdot u[n] \xrightarrow{z} \frac{1 \cdot z^{-1}}{(1-z^{-1})^2} \rightarrow (1-z^{-1})^2 = 0 \Rightarrow z=1 \quad |z| > 1$$

3. $u[n-N]$, brug (4).

$$(\delta[n-m]) \xrightarrow{z} z^{-m}$$

$$u[n-N] \xrightarrow{z} z^{-N}$$

Nu samler vi det hele:

$$\left(\frac{z^{-1}}{(1-z^{-1})^2} \right) - \left(\frac{z^{-1}}{(1-z^{-1})^2} \right) \cdot (z^{-N})$$

$$X(z) = \frac{z^{-1} - z^{-N-1}}{(1-z^{-1})^2} = \frac{z^{-1}(1-z^{-N})}{(1-z^{-1})^2}$$

d) $x[n] = a^n \quad 0 \leq |a| < 1$

$$a^n = \begin{cases} a^n & n \geq 0 \\ a^{-n} & n < 0 \end{cases}$$

$$X(z) = \sum_{n=-\infty}^{-1} a^{-n} z^{-n} + \sum_{n=0}^{\infty} a^n z^{-n} = \sum_{n=1}^{\infty} a^n z^n + \sum_{n=0}^{\infty} a^n z^{-n}$$

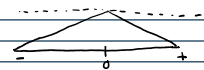
$$(1) \sum_{n=1}^{\infty} (az)^n = \sum_{n=0}^{\infty} (az)^n - (az)^0 = \frac{1}{1-az} - 1 = \frac{az}{1-az}$$

$$(2) \sum_{n=0}^{\infty} (az^{-1})^n = \frac{1}{1-az^{-1}}$$

We therefore obtain

$$X(z) = \frac{az}{1-az} + \frac{1}{1-az^{-1}} \quad |a| < |z| < \frac{1}{|a|}$$

Splittet den op i $-\infty \rightarrow -1$ & $0 \rightarrow +\infty$



Flipper den for at ikke have "de negative freq."

$$\frac{1}{1-az} - 1 = \frac{1}{1-az} - \frac{1-az}{1-az} = \frac{1-(1-az)}{1-az} = \frac{1-1+az}{1-az} = \frac{az}{1-az}$$

Centrere den ved 0

$$1-az = 0 \Rightarrow z = a$$

$$1-az^{-1} = 0 \Rightarrow z = a$$

$$h) \quad x[n] = \begin{cases} 1 & 0 \leq n < N-1 \\ 0 & n \geq N \\ 0 & n < 0 \end{cases}$$

$$\begin{aligned} X(z) &= \sum_{n=0}^{N-1} z^{-n} = \sum_{n=0}^{\infty} (z^{-1})^n - \sum_{n=N}^{\infty} (z^{-1})^n = \sum_{n=0}^{\infty} (z^{-1})^n - z^{-N} \sum_{n=0}^{\infty} (z^{-1})^n \\ &= \sum_{n=0}^{\infty} (z^{-1})^n (1 - z^{-N}) = \frac{1 - z^{-N}}{1 - z^{-1}} \\ &= \frac{1 - \frac{1}{z^N}}{1 - \frac{1}{z}} = \frac{\frac{z^N - 1}{z^N}}{\frac{z - 1}{z}} = \frac{z^N - 1}{z^{N-1}(z - 1)} \end{aligned}$$

Note that the pole in 1 is canceled by the zero in 1.
Therefore the series converges for $z \neq 0$